THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **Patterns in Mathematics** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 81	First term + (12-1) x 7 = 4+11x7
2	**(c)** 18	Number of steps from the first term is $(\text{term} - \text{first}) \div d$; the term number is one more.
3	**(a)** 156	Sum of first n even numbers is $n \times (n+1)$.
4	**(c)** 210	Add 20.
5	**(a)** 21	Sticks $= 3 + 2 \times (n-1) = 2n+1$.
6	**(d)** 625	Consecutive triangular numbers add to a square.
7	**(d)** 128	Each term is double the one before it.
8	**(c)** 93	First term + (15-1) x 6 = 9+14x6
9	**(b)** 25	Number of steps from the first term is $(\text{term} - \text{first}) \div d$; the term number is one more.
10	**(c)** 240	Sum of first n even numbers is $n \times (n+1)$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Rule: 20th term = first term $+(20-1)\times$ difference $= 5 + 19 \times 3$ (1); 20th term $= 62$ (1).	1+1
12	Terms are 3, 7, ... each 4 more than the last (1); 35 is a term but 36 lies between two terms, so it is not a term (1).	1+1
13	Odd: $30^2 = 900$ (1); Even: $30 \times 31 = 930$ (1).	1+1
14	Rule: 15th term = first term $+(15-1)\times$ difference $= 2 + 14 \times 7$ (1); 15th term $= 100$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sticks for 12 triangles $= 3 + 2 \times (12-1) = 25$ (1); For 35 sticks: $3 + 2 \times (m-1) = 35$, so $2 \times (m-1) = 32$ (1); $m - 1 = 16$, so 17 triangles (1).	1+1+1
16	There are 30 odd numbers, sum $= 900$ (1); Even numbers: sum $= 30 \times 31$ (1); $= 930$ (1).	1+1+1
17	$T_{20} = 210$, $T_{21} = 231$ (1); Sum $= 441$ (1); $21^2 = 441$ matches (1).	1+1+1
	OR: Sticks for 15 triangles $= 3 + 2 \times (15-1) = 31$ (1); For 41 sticks: $3 + 2 \times (m-1) = 41$, so $2 \times (m-1) = 38$ (1); $m - 1 = 19$, so 20 triangles (1).	1+1+1
18	There are 40 odd numbers, sum $= 1600$ (1); Even numbers: sum $= 40 \times 41$ (1); $= 1640$ (1).	1+1+1
	OR: $T_{24} = 300$, $T_{25} = 325$ (1); Sum $= 625$ (1); $25^2 = 625$ matches (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	10th figure $= 5 + (10-1) \times 4 = 41$ (1); More than 91 needs $(91-5) \div 4$ more steps (1); Steps needed: 13 (1); Figure number 23 (1); Check by substituting (1).	1+1+1+1+1
	OR: Sum of first 9 numbers $= \frac{9 \times 10}{2} = 45$ (1); Sum of cubes $= 2025$ (1); $45^2 = 2025$ (1); They are equal, so the cube sum is a perfect square (2).	1+1+1+2
20	12th figure $= 3 + (12-1) \times 5 = 58$ (1); More than 108 needs $(108-3) \div 5$ more steps (1); Steps needed: 11 (1); Figure number 23 (1); Check by substituting (1).	1+1+1+1+1
	OR: Sum of first 10 numbers $= \frac{10 \times 11}{2} = 55$ (1); Sum of cubes $= 3025$ (1); $55^2 = 3025$ (1); They are equal, so the cube sum is a perfect square (2).	1+1+1+2

Mistakes to avoid:

- Thinking the sum of the first n natural numbers is $n \times n$; that is the sum of the first n **odd** numbers.
- Guessing a rule from only two terms. Always check the rule on at least three terms.
- Mixing up the nth odd number $2n-1$ with the nth even number $2n$.

- Treating $2, 4, 8, 16$ as having a common difference; the gaps are $2, 4, 8$, so no constant is added.
- In the Virahanka sequence, adding the wrong two terms (the rule uses the *two terms just before*).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **Patterns in Mathematics** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 300	Add 24.
2	**(a)** 31	Sticks $= 3 + 2 \times (n - 1) = 2n + 1$.
3	**(a)** 900	Consecutive triangular numbers add to a square.
4	**(c)** 256	Each term is double the one before it.
5	**(c)** 93	First term + (11-1) x 8 = 13+10x8
6	**(b)** 13	Number of steps from the first term is $(\text{term} - \text{first}) \div d$; the term number is one more.
7	**(c)** 342	Sum of first n even numbers is $n \times (n + 1)$.
8	**(c)** 406	Add 28.
9	**(b)** 41	Sticks $= 3 + 2 \times (n - 1) = 2n + 1$.
10	**(d)** 1225	Consecutive triangular numbers add to a square.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Terms are 6, 11, ... each 5 more than the last (1); 61 is a term but 62 lies between two terms, so it is not a term (1).	1+1
12	Odd: $35^2 = 1225$ (1); Even: $35 \times 36 = 1260$ (1).	1+1
13	Rule: 18th term = first term $+(18 - 1)\times$ difference $= 11 + 17 \times 4$ (1); 18th term $= 79$ (1).	1+1
14	Terms are 2, 8, ... each 6 more than the last (1); 80 is a term but 81 lies between two terms, so it is not a term (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sticks for 18 triangles $= 3 + 2 \times (18 - 1) = 37$ (1); For 47 sticks: $3 + 2 \times (m - 1) = 47$, so $2 \times (m - 1) = 44$ (1); $m - 1 = 22$, so 23 triangles (1).	1+1+1
16	There are 45 odd numbers, sum $= 2025$ (1); Even numbers: sum $= 45 \times 46$ (1); $= 2070$ (1).	1+1+1
17	$T_{28} = 406$, $T_{29} = 435$ (1); Sum $= 841$ (1); $29^2 = 841$ matches (1).	1+1+1
	OR: Sticks for 21 triangles $= 3 + 2 \times (21 - 1) = 43$ (1); For 53 sticks: $3 + 2 \times (m - 1) = 53$, so $2 \times (m - 1) = 50$ (1); $m - 1 = 25$, so 26 triangles (1).	1+1+1
18	There are 50 odd numbers, sum $= 2500$ (1); Even numbers: sum $= 50 \times 51$ (1); $= 2550$ (1).	1+1+1
	OR: $T_{32} = 528$, $T_{33} = 561$ (1); Sum $= 1089$ (1); $33^2 = 1089$ matches (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	9th figure $= 8 + (9 - 1) \times 6 = 56$ (1); More than 106 needs $(106 - 8) \div 6$ more steps (1); Steps needed: 9 (1); Figure number 18 (1); Check by substituting (1).	1+1+1+1+1
	OR: Sum of first 11 numbers $= \frac{11 \times 12}{2} = 66$ (1); Sum of cubes $= 4356$ (1); $66^2 = 4356$ (1); They are equal, so the cube sum is a perfect square (2).	1+1+1+2
20	11th figure $= 4 + (11 - 1) \times 7 = 74$ (1); More than 124 needs $(124 - 4) \div 7$ more steps (1); Steps needed: 8 (1); Figure number 19 (1); Check by substituting (1).	1+1+1+1+1
	OR: Sum of first 12 numbers $= \frac{12 \times 13}{2} = 78$ (1); Sum of cubes $= 6084$ (1); $78^2 = 6084$ (1); They are equal, so the cube sum is a perfect square (2).	1+1+1+2

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- Guessing a rule from only two terms. Always check the rule on at least three terms.
- Mixing up the nth odd number $2n - 1$ with the nth even number $2n$.
- Treating $2, 4, 8, 16$ as having a common difference; the gaps are $2, 4, 8$, so no constant is added.

- In the Virahanka sequence, adding the wrong two terms (the rule uses the *two terms just before*).

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **Patterns in Mathematics** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 512	Each term is double the one before it.
2	**(d)** 122	First term + (14-1) x 9 = 5+13x9
3	**(a)** 22	Number of steps from the first term is $(\text{term} - \text{first}) \div d$; the term number is one more.
4	**(a)** 506	Sum of first n even numbers is $n \times (n+1)$.
5	**(b)** 528	Add 32.
6	**(b)** 51	Sticks $= 3 + 2 \times (n-1) = 2n+1$.
7	**(d)** 1600	Consecutive triangular numbers add to a square.
8	**(b)** 1024	Each term is double the one before it.
9	**(c)** 104	First term + (9-1) x 11 = 16+8x11
10	**(b)** 31	Number of steps from the first term is $(\text{term} - \text{first}) \div d$; the term number is one more.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Odd: $40^2 = 1600$ (1); Even: $40 \times 41 = 1640$ (1).	1+1
12	Rule: 25th term = first term $+(25-1)\times$ difference $= 6 + 24 \times 5$ (1); 25th term $= 126$ (1).	1+1
13	Terms are 9, 17, ... each 8 more than the last (1); 81 is a term but 82 lies between two terms, so it is not a term (1).	1+1
14	Odd: $50^2 = 2500$ (1); Even: $50 \times 51 = 2550$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sticks for 24 triangles $= 3 + 2 \times (24-1) = 49$ (1); For 59 sticks: $3 + 2 \times (m-1) = 59$, so $2 \times (m-1) = 56$ (1); $m - 1 = 28$, so 29 triangles (1).	1+1+1
16	There are 60 odd numbers, sum $= 3600$ (1); Even numbers: sum $= 60 \times 61$ (1); $= 3660$ (1).	1+1+1
17	$T_{36} = 666$, $T_{37} = 703$ (1); Sum $= 1369$ (1); $37^2 = 1369$ matches (1).	1+1+1
	OR: Sticks for 27 triangles $= 3 + 2 \times (27-1) = 55$ (1); For 65 sticks: $3 + 2 \times (m-1) = 65$, so $2 \times (m-1) = 62$ (1); $m - 1 = 31$, so 32 triangles (1).	1+1+1
18	There are 75 odd numbers, sum $= 5625$ (1); Even numbers: sum $= 75 \times 76$ (1); $= 5700$ (1).	1+1+1
	OR: $T_{40} = 820$, $T_{41} = 861$ (1); Sum $= 1681$ (1); $41^2 = 1681$ matches (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	14th figure $= 6 + (14-1) \times 3 = 45$ (1); More than 95 needs $(95-6) \div 3$ more steps (1); Steps needed: 17 (1); Figure number 31 (1); Check by substituting (1).	1+1+1+1+1
	OR: Sum of first 13 numbers $= \frac{13 \times 14}{2} = 91$ (1); Sum of cubes $= 8281$ (1); $91^2 = 8281$ (1); They are equal, so the cube sum is a perfect square (2).	1+1+1+2
20	8th figure $= 2 + (8-1) \times 8 = 58$ (1); More than 108 needs $(108-2) \div 8$ more steps (1); Steps needed: 7 (1); Figure number 15 (1); Check by substituting (1).	1+1+1+1+1
	OR: Sum of first 14 numbers $= \frac{14 \times 15}{2} = 105$ (1); Sum of cubes $= 11025$ (1); $105^2 = 11025$ (1); They are equal, so the cube sum is a perfect square (2).	1+1+1+2

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